Measurement of Mode Field Profiles and Bending and Transition Losses in Curved Optical Channel Waveguides

Vijaya Subramaniam, Gregory N. De Brabander, David H. Naghski, *Student Member, IEEE,* and Joseph T. Boyd, *Senior Member, IEEE*

***Abstract*—Curved, single-mode, silicon oxynitride optical ridge channel waveguides have been characterized by measuring mode field profile alteration, bending loss, and transition loss. The results were compared to theoretical calculations. Near field imaging of the channel mode field profiles showed the effect of bend radius on mode shape. Good agreement was obtained between the measurements and profiles obtained from two-dimensional (2-D) beam propagation method calculations. The exponential dependence of bending loss on bend radius and the variation of transition loss on the lateral offset between channels having different bend radius were successfully modeled by two simple theories.**

***Index Terms*—Integrated optics, optical losses, optical propagation, optical waveguides, optical waveguide theory, ridge waveguides, waveguide bends.**

I. Introduction

OPTICAL channel waveguides are an important component of photonic integrated circuits [1]–[3]. Curved channel waveguides are used to interconnect noncollinear straight channels and, in addition to the loss mechanisms encountered in straight waveguides, curved channels incur losses due to bending and transition. For practical photonic integrated circuits it is not only important to keep these excess losses small, but to also make compact curves, thereby allowing the highest degree of integration possible. Successful integrated optic design demands that one be able to accurately estimate the expected loss of the component waveguides in order to optimize the throughput of a given device.

In a straight channel waveguide, the optical field is symmetric about the field peak which occurs at the center of the channel. For the lowest order mode the field profile varies approximately in a sinusoidal manner between the sides of the channel and approximately exponentially beyond. By contrast, the optical field in a curved waveguide having the same cross section has the peak displaced laterally from the channel center in a direction away from the arc center. The mode field is also deformed in that it is asymmetric and has a different width from that of the straight guide.

Bending loss is the continuous radiation of mode power tangentially out of a curved waveguide as light travels around the bend. This radiation occurs because the mode field of a curved guide is leaky, a fact which can be shown through the technique of conformal mapping [4]. Bending loss can be reduced by making the bend radius larger or by increasing the confinement of the mode field. However, these changes result in either an increase in overall device length or increased insertion loss. Transition loss occurs at the junction of two waveguides having different radii of curvature. The mode fields of the waveguides will be mismatched, due to differing degrees of mode field peak shift and deformation, causing a reduction in the power transfer between the two waveguides. This loss may be reduced by changing the curvature in a gradual manner or by laterally offsetting the waveguide sections so as to reduce mode mismatch.

Theoretical descriptions of these loss mechanisms in both fibers and waveguides have been the source of continuous study. For design purposes, closed form analytic equations are simpler to implement for loss estimation. This has been accomplished using two-dimensional (2-D) models, where the fiber or waveguide is transformed to an equivalent planar waveguide using the effective index method [5]–[10]. The resulting equations have been found to adequately describe the experimental results of bending loss measurements [11], [12] and transition loss measurements [13] for weakly guiding configurations. Improvements in the accuracy of these solutions have been obtained but often at the expense of no longer obtaining closed-form solutions [4], [14]–[17]. However, at least one study has shown that measurement uncertainties may invalidate the increased accuracy of more complicated analyzes [18].

Three-dimensional (3-D) analyses have been developed which attempt to more rigorously account for the index profile present in rib-channel waveguides. These analyzes may be necessary when the effective index method is rendered unusable by the implementation of stronger optical confinement, a situation which occurs with deeply etched waveguides or when using materials with a large index difference between the guiding and cladding layers, such as semiconductors [19],

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V. Subramaniam was with the Department of Electrical and Computer Engineering and Computer Science, University of Cincinnati, Cincinnati, OH 45221 USA. She is now with VLSI Technology, Inc., Austin, TX 78712 USA.

G. N. De Brabander was with the Department of Electrical and Computer Engineering and Computer Science, University of Cincinnati, Cincinnati, OH 45221 USA. He is now with Hewlett-Packard, Inc., San Jose, CA 95114 USA.

D. H. Naghski and J. T. Boyd are with the Department of Electrical and Computer Engineering and Computer Science, University of Cincinnati, Cincinnati, OH 45221 USA.

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[20]. Losses calculated numerically using these analyzes have agreed well with experimental results [19]–[24] but the fact that closed form solutions are not obtained is often a drawback from a design point of view. The computations for loss are numerical and often the field profile is needed, which in itself entails an additional numerical procedure.

We have conducted measurements on rib channel waveguides formed into U-bend, S-bend, and straight sections. The mode field profiles of straight and curved channel waveguides have been measured using near field imaging. The profiles of the curved guides clearly show the deformation of the field due to propagation through the bend. To our knowledge, these measurements have not been performed on curved channel waveguides although similar measurements have been performed on single mode fibers [25]. Comparisons are made to profiles calculated using either a radially dependent Gaussian model or the beam propagation method (BPM). We have also measured bending and transition losses and compare the results to expected loss values calculated using analytic equations derived from two models, as well as to BPM calculations. The analytic equations are simple, closed form, easy to implement, and are found to be accurate for the weakly guiding waveguides we have used.

II. Design Considerations and Fabrication

The waveguides fabricated in this study consisted of a silicon oxynitride (SiON) waveguiding layer deposited on a lower cladding of silicon dioxide which was thermally grown on a $\langle 100 \rangle$ silicon substrate. Details of the fabrication procedure have been published previously [26]. Channel waveguides of interest were U-bends, S-bends, and straight guides, as shown in Fig 1. The waveguide cross section is shown in Fig. 2(a). The effective index method [27] was used to model and direct the design of the rib channel waveguide sample. The equivalent planar waveguide is shown in Fig. 2(b) with two sets of effective index values. The values of n_{e1} and n_{e2} not in parenthesis were calculated from the refractive index values given in Fig. 2(a) while those in parenthesis were measured by prism coupling as described below. Values of Δn_e and n_e without and within parenthesis were calculated from corresponding values of n_{e1} and n_{e2}. The guide layer thickness and refractive index, the etch depth, and the waveguide width were chosen to provide single mode operation at a wavelength of 830 nm. The thickness of the isolation layer was set to give a negligible TE mode substrate coupling loss of approximately 0.1 dB/cm. The values of isolation layer thickness, waveguide width, and etch depth were measured by stylus profilometry. The SiON thickness and refractive index were determined both by prism coupling measurements on waveguide samples and ellipsometrically on a monitor substrate. Established values were utilized for the refractive indexes of SiO_2 and Si.

The bending loss expected in the curved waveguides was also a part of the design process and was estimated by means of the commonly used model of Marcuse [6]. However, the magnitude of this loss is critically dependent on the difference between the effective indexes of the etched and unetched

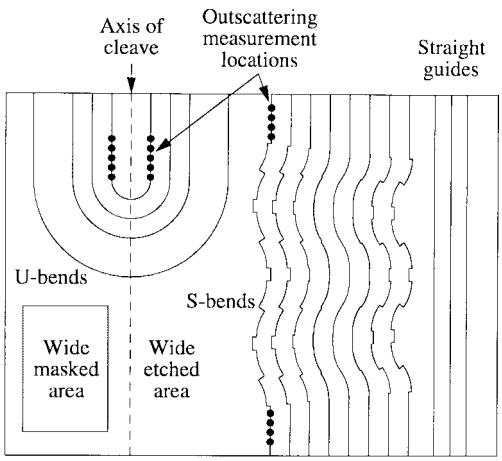


Fig. 1. Schematic of sample layout showing U-bends, S-bends, and straight channel waveguides. The wide masked areas used for prism coupling as well as out-scattering measurement locations are indicated. Also shown is the position of the cleave exposing the end face where arc field profiles were measured.

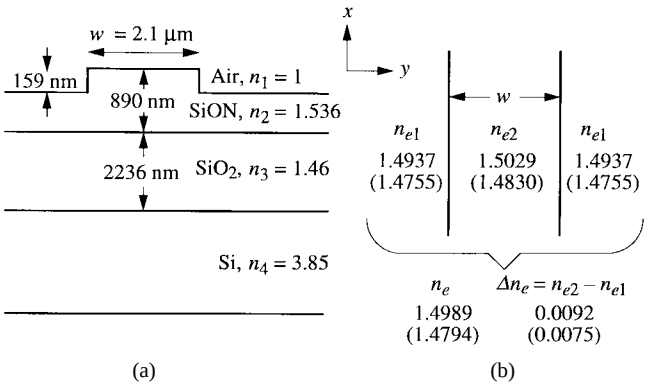


Fig. 2. (a) Cross-section of waveguide sample. (b) Effective index model equivalent planar waveguide.

areas, Δn_e. Because of uncertainties in the ellipsometrically measured refractive index of the SiON layer (± 0.02) the TE effective refractive indexes of etched or unetched regions, n_{e1} or n_{e2}, respectively, were measured by prism coupling into wide planar guiding areas on the sample (Fig. 1). These values, given in parenthesis in Fig. 2(b), are believed to be more accurate than the values calculated from the nominal material indexes and were used in all subsequent model calculations.

III. Theory

We have used two theoretical analyzes to calculate both the expected bending loss and the expected transition loss for our waveguide sample. Each uses the effective index approximation to reduce the two dimensional index profile to a one dimensional planar waveguide. Each analysis results in a closed form analytic expression for loss which may easily be evaluated.

For a bent slab waveguide, Marcuse obtained an equation for the exponential power attenuation coefficient due to bending loss using a rigorous analysis of the wave equation in cylindrical coordinates [6]. Using effective index notation, this

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attenuation coefficient α_{bend} is

$$\alpha_{\text{bend}} = \frac{\alpha_y^2}{k_0^3 n_e\left(1+\alpha_y\frac{w}{2}\right)}\frac{k_y^2}{(n_{e2}^2-n_{e1}^2)}\exp(\alpha_y w) \cdot \exp\left(\frac{-2\alpha_y^3}{3n_e^2k_0^2}R\right) \tag{1}$$

where

$$\alpha_y = k_0\sqrt{n_e^2-n_{e1}^2}, \quad k_y = k_0\sqrt{n_{e2}^2-n_e^2}, \quad \text{and} \quad k_0 = 2\pi/\lambda_0.$$

The transmission through a bend of radius of curvature R and included angle θ is

$$\eta_{\text{bend}} = \exp(-\alpha_{\text{bend}}\theta R) \tag{2}$$

and the bending loss in dB is then

$$L_{\text{bend}} = -10\log_{10}(\eta_{\text{bend}}) = (4.34)\alpha_{\text{bend}}\theta R. \tag{3}$$

The transmission that takes place at a radius of curvature discontinuity is obtained by computing the overlap between the fields of two waveguides bent in opposite directions. To approximate the field in a curved waveguide we have adapted the fiber analysis of Gambling *et al*. [10], which gives the electric field profile, profile peak displacement, and transition loss in a curved fiber. To use this theory we first assume that the electric field profile in a straight waveguide can be approximated by a Gaussian

$$E_0(y) = A\exp\left[-\frac{1}{2}\left(\frac{y}{\omega_0}\right)^2\right] \tag{4}$$

with $y=0$ at the guide center. A radial dependence (y) is added to give the field in a curved waveguide, with radius of curvature R as

$$E(y) = E_0(y)\left[1+\frac{1}{R}(k_0 n_{e2}\omega_0)^2 y\right] \tag{5}$$

where $E_0(y)$ is given by (4) and y increases as one moves away from the center of the arc. The mode width ω_0 is found by fitting (4) to the field profile of the waveguide as determined by the effective index method. For our sample, using the measured values of n_{e1} and n_{e2} and fitting over an 8 μm range, $|y| \le 4\,\mu\text{m}, \omega_0 = 1.37\,\mu\text{m}$ was obtained. The value of ω_0 did not vary by more than 1% for any fit range covering at least 4 μm (centered at $y=0$).

Two waveguides bent in opposite directions, and having radii of curvature R_1 and R_2 and positive offset a are shown in Fig. 3. The curve-to-curve transmission η_{cc} between these guides is determined from the overlap integral

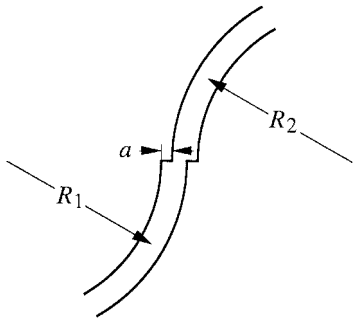


Fig. 3. Junction of two waveguide arcs showing a positive offset $a>0$.

$$\eta(a) = \frac{\left|\int_{-\infty}^{\infty} E_1(y)E_2^*(a-y)\,dy\right|^2}{\int_{-\infty}^{\infty}E_1(y)E_1^*(y)\,dy\int_{-\infty}^{\infty}E_2(y)E_2^*(y)\,dy} \tag{6}$$

where E_1 and E_2 are found using (5) with $R=R_1$ and $R=R_2$, respectively. Inserting (5) into (6), and simplifying, gives (7) shown at the bottom of the page.

The transmission at a straight-to-curve transition η_{sc} is found from (7) by letting $R_1 \to \infty$ and substituting $R_2 = R$ to obtain

$$\eta_{sc}(a) = \frac{\exp\left(-\frac{a^2}{2\omega_0^2}\right)\left[1+\frac{(k_0 n_{e2}\omega_0)^2 a}{2R}\right]^2}{\left[1+\frac{(k_0 n_{e2}\omega_0)^4\omega_0^2}{2R^2}\right]}. \tag{8}$$

Thus, (7) and (8) give closed-form equations for the transmission that are straightforward to evaluate. The losses at these transitions in dB are

$$L_{cc} = -10\log_{10}(\eta_{cc}) \quad \text{and} \quad L_{sc} = -10\log_{10}(\eta_{sc}). \tag{9}$$

The desired offset at either of these transitions is that which maximizes the transmission, found by solving $d\eta/da = 0$ for a_{opt}. Although analytic expressions for a_{opt} are found for both η_{sc} and η_{cc}, the expression for η_{cc} is quite complex. Alternatively, to a first approximation, it can be assumed that the loss between waveguides of different radii will be minimized when they are offset to make their field peaks coincide. The electric field peak displacement $d(R)$ is found by setting $dE/dy = 0$ and solving for y, giving

$$d(R) = \frac{\sqrt{1+\frac{4(k_0 n_{e2}\omega_0)^4\omega_0^2}{R^2}}-1}{\frac{2}{R}(k_0 n_{e2}\omega_0)^2}. \tag{10}$$

$$\eta_{cc}(a) = \frac{\exp\left(-\frac{a^2}{2\omega_0^2}\right)\left\{\left[1+\frac{k_0 n_{e2}\omega_0)^2 a}{2R_1}\right]\left[1+\frac{(k_0 n_{e2}\omega_0)^2 a}{2R_2}\right]-\frac{(k_0 n_{e2}\omega_0)^4\omega_0^2}{2R_1R_2}\right\}^2}{\left[1+\frac{(k_0 n_{e2}\omega_0)^4\omega_0^2}{2R_1^2}\right]\left[1+\frac{(k_0 n_{e2}\omega_0)^4\omega_0^2}{2R_2^2}\right]}. \tag{7}$$

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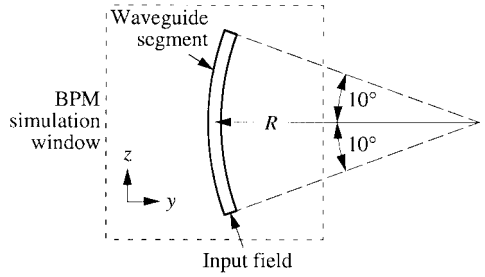


Fig. 4. Schematic illustration of the BPM simulation model for curved guides.

For $R \gg (k_0 n_{e2} \omega_0)^2 \omega_0$ (10) simplifies further to

$$\tilde{d}(R) \approx \frac{(n_{e2} k_0 \omega_0)^2 \omega_0^2}{R}. \tag{11}$$

Comparisons were made between the transmission calculated for straight-to-curve transitions using offsets of $a = \tilde{d}(R)$ and $a = a_{\text{opt}}$. These showed that for our waveguide cross section, for radii $R \geq 1000\,\mu$m, $\eta_{sc}(a = \tilde{d}(R))$ differed from $\eta_{sc}(a = a_{\text{opt}})$ by at most 1%. Similar calculations using $a = 2\tilde{d}(R)$ and $a = a_{\text{opt}}$ for curve-to-curve transitions with $R_1 = R_2 = R$, showed a maximum difference of only 4% between the approximate and exact values. Therefore, from the standpoint of device design, (11) may be used for offset calculation.

Results calculated from (3) and (9) were compared to those obtained with a commercial beam propagation method program [28]. The 2-D implementation of the program was used to determine the complex field amplitude profile of the curved guides. Fig. 4 illustrates how a curved waveguide was modeled as an arc of radius R with an included angle of 20°. The BPM program limits nonparaxiality errors by utilizing a (3, 3) Padé approximation to the standard parabolic approximation of the Helmholtz equation. In order to further reduce nonparaxial effects, the chord connecting the endpoints of the arc was set parallel to the z axis of the simulation window and the mode of the straight guide, with phase front tilted to 10°, was launched into the arc. The included angle was chosen to allow sufficient propagation distance for the input to equilibrate to the mode of the curved guide. For each radius simulated, negligible difference was found between the field amplitude distributions at the middle and at the end of the arc, showing that by the middle of the arc the input mode had become that of the curved waveguide. Bending loss, in dB/degree of arc length, was computed directly by the program from decay of mode power. Transition loss was calculated by numerical evaluation of the overlap between the complex fields of either straight and curved or curved and curved waveguides. The simulation parameters were free space wavelength $\lambda_o = 0.83\,\mu$m, cladding index 1.4755, and $\Delta n_e = 0.0075$. The computation mesh was $\Delta y = 0.05\,\mu$m and $\Delta z = 0.1\,\mu$m.

IV. Loss Measurements

The out-scattering technique, which measures the intensity of the out-scattered light from the channel waveguide as a function of distance from the incoupling point, was used to determine propagation loss [2], [29]. Polarized light from an 830 nm laser diode was collimated and focused onto the cleaved edge of the channel waveguide to excite the quasi-TE mode. The outscattered light at a particular location on the channel is assumed to be proportional to the amount of light passing through that portion of the channel and the proportionality constant and loss is assumed to remain uniform along the channel. The transmission at any point is then

$$\eta(\ell) = \frac{I_{\text{out}}(\ell)}{I_{\text{in}}} = \exp(-\alpha \ell) \tag{12}$$

while the normalized loss in dB/cm (ℓ in cm) is

$$L = \frac{-10 \log_{10}[\eta(\ell)]}{\ell} = (4.34)\alpha \tag{13}$$

The outscattered light was measured with a set-up similar to that used in a previous study [30]. Here, a 25× long working distance objective was moved parallel to the channel, imaging the waveguide surface onto a fiber bundle, which carried the surface light to a photomultiplier tube.

The background loss of the sample, which encompasses all loss mechanisms other than bending and transition (e.g., scattering and absorption) and is assumed to be representative of all waveguides, was measured first. The background attenuation coefficient α_{back} was determined by performing out-scattering measurements along a straight channel and, using the definitions of (12) and (13), finding the slope of a plot of the $\log_{10}$ of transmission as a function of distance. In order to remove effects of sample nonuniformity, this measurement was performed twice, coupling into each end of the channel. The average of these two measurements gave $L_{\text{back}} = 1.24$ dB/cm. The background loss was subtracted from all loss values subsequently measured on the U-bend and S-bend structures to give "corrected measured" values as

$$L_{\text{corrected measured}} = L_{\text{measured}} - \ell L_{\text{back}} \tag{14}$$

where ℓ is the path length between measurement points. These values then consisted of losses due to bends and transitions only.

The U-bends were concentric 180° arcs, with radius of curvature ranging from 100 to 4500 μm in steps of 100 μm where each end of an arc was connected to the sample edge by tangential input and output straight sections. Loss measurements were carried out for U-bends of bend radius ranging from 1900 to 4500 μm. Scattering intensity measurements were taken at five points, each separated by 0.1 mm, on both the input and output straight sections of each U-bend near where the bend begins, as shown in Fig. 1. The measured input and output intensities were taken as the average of the respective five measurements. This resultant loss was corrected using (14) for background loss over the length of the bend $\ell = \pi R$. The small loss of the straight sections between the group of measurement points and the bend was ignored. The theoretical loss of the U-bend, caused by bending loss and the

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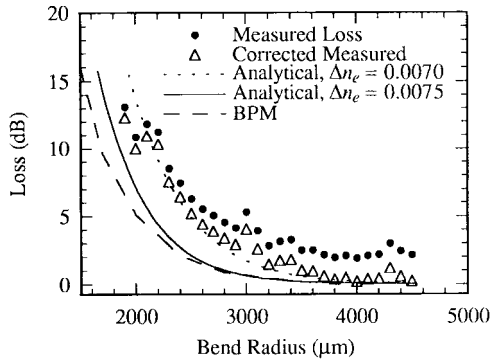


Fig. 5. Comparison of measured loss, corrected measured loss (straight guide propagation loss is subtracted), analytical loss [using (1)], and BPM loss in U-bends. The difference between the two analytical curves illustrates the sensitivity of (1) to changes in Δn_e.

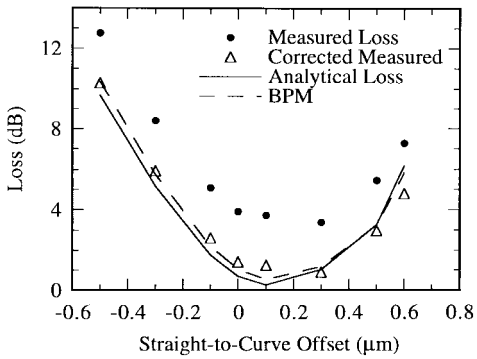


Fig. 6. Comparison of measured loss, corrected measured loss (straight guide propagation loss is subtracted), analytical loss [using (7) and (8)], and BPM loss in S-bends.

two straight-to-curve transitions, was calculated as

$$L_{\text{U-bend}} = L_{\text{bend}} + 2L_{sc} \tag{15}$$

using the appropriate radius in (1) and (8), with an included angle of $\theta = 180°$, and with $a = 0$.

Measured and corrected measured loss as a function of U-bend radius as well as losses calculated analytically and using BPM are shown in Fig. 5. Two analytical curves are shown in Fig. 5 in order to illustrate the critical dependence of bending loss (1) on the value of Δn_e. The bending loss is underestimated by (1) using the measured Δn_e of 0.0075, while for a Δn_e only 0.0005 smaller, good agreement is obtained. This difference in Δn_e would occur if the refractive index of the SiON film, where the guides are located, was 0.005 higher (0.3%) than where the prism measurements were taken or if the difference in the measured prism coupling angles was too low by 0.005°. Therefore, the actual values of material parameters have a very strong influence on the success of design modeling. When comparing the analytical model (using $\Delta n_e = 0.0075$) to the BPM results, it is seen that there is good agreement for radii greater than 2000 μm which is the most likely design regime because of the low loss. Although the analytical model begins to significantly overestimate the loss for smaller radii, these radii would probably not be used due to the high loss.

S-bends, designed to test the effects of transition loss, each consisted of eight straight-to-curve transitions and four curve-to-curve transitions, as illustrated in Fig. 1. The S-bends were formed using eight arc segments, each having a radius of 3500 μm and an included angle of 35°, and three connecting straight channels, each 1 mm long. There were a total of eight S-bends of equal path length, four with positive-offset, three with negative-offset, and one with zero offset, at the transitions. Positive offset $(a > 0)$ displaces the guides such that the peaks of the fields in the waveguide sections are brought closer, thereby increasing overlap and reducing loss. However large values of positive offset will overdisplace the fields leading to higher loss. Negative offset $(a < 0)$ displaces the guides such that the peaks of the fields are shifted away from one another, leading to less overlap and hence more loss. For the eight S-Bends, the straight-to-curve offsets in microns were −0.5, −0.3, −0.1, 0, 0.1, 0.3, 0.5, 0.6 with corresponding curve-to-curve offsets of −0.86, −0.58, −0.22, 0, 0.22, 0.58, 0.86, 1.16, respectively. This wide variety of offset values, including negative offsets which always increase transition loss, allows the loss minimum to be clearly delineated. All curve-to-curve offset values are for a lateral offset perpendicular to the direction of the guide where the transition occurs and the endpoints are still on the 0.1 μm e-beam photomask grid.

Loss measurements were carried out on the S-bends in a manner similar to that used on the U-bends. Four readings were taken on both the input and output straight channels at intervals of 0.1 mm, as indicated in Fig. 1, with the respective averages yielding the input and output intensity. Equation (14) was used to determine corrected measured loss by subtracting off the straight channel propagation loss for $\ell = 2.01$ cm, the path length between the input and output measurement positions. The loss caused by bending loss in the four arc segments, and transition loss in the eight straight-to-curve transitions and the four curve-to-curve transitions was calculated analytically as

$$L_{\text{S-bend}} = 4L_{\text{bend}} + 8L_{sc} + 4L_{cc}. \tag{16}$$

The analytical bending loss for each curved segment is obtained using (1) and (3) with $R = 3500\,\mu$m and $\theta = 35°$ ($L_{\text{bend}} = 0.031$ dB per segment). The analytical transition loss was obtained by substituting the appropriate value of offset for the straight-to-curve and curve-to-curve transitions in (7) and (8), respectively.

A plot of measured, corrected measured, analytical, and BPM loss for the S-bends as a function of straight-to-curve offset is shown in Fig. 6. The latter three sets of values show good agreement. The transition loss is a minimum for a positive offset bend. The slight leftward shift of the analytical curve away from the BPM curve suggests that the minima of the curves do not occur at the same offset value. Although the actual measured minimum loss, obtained for a bend having a positive straight-to-curve offset of 0.3 μm and a curve-to-curve offset of 0.58 μm, does not occur exactly at either the analytical or BPM minima, the difference is well within the loss measurement variation.

Fig. 7 shows the field peak displacement calculated by (10) and (11) and that determined by BPM. The approximation (11) agrees with (10) except toward small bend radii. However, the

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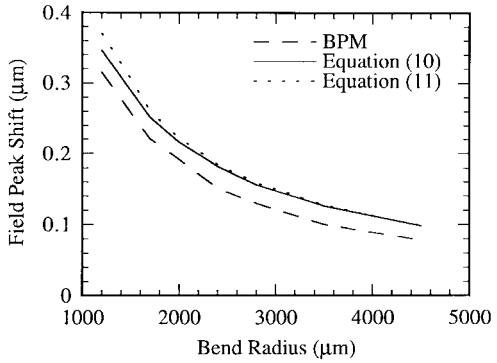


Fig. 7. Field peak displacement as a function of bend radius calculated by (10), (11), and determined by BPM.

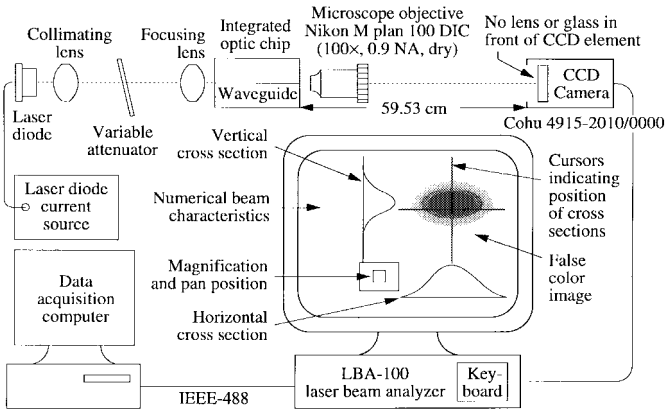


Fig. 8. Experimental setup for mode intensity measurement in waveguides.

field peak displacement calculated by either (10) or (11) is 25% more than that determined by BPM. Despite this seemingly large overestimation of the optimum offset value, a design based on (11) would incur a negligible loss penalty due to the small variation in loss in the vicinity of the optimum offset.

V. Mode Field Profile Measurement

The mode intensity profiles of the straight and curved channel waveguides were measured using a Spiricon^{TM} laser beam analyzer. As shown schematically in Fig. 8, light polarized to excite the quasi-TE mode from an 830 nm laser diode was collimated and focused onto the cleaved input edge of the channel waveguide. A microscope objective was used to image light emerging from the cleaved end of the waveguide onto a windowless silicon charged-coupled device (CCD) camera. The mode field profiles of curved waveguides were measured after cleaving the sample through the center of the concentric U-bend arcs perpendicularly to the edge where the straight lead-in channels terminated (Fig. 1). Intensity profile measurements were made on both halves of the sample on a number of bends of different radii at the cleaved edge where the arcs were interrupted. Mirror image results were obtained for left and right-handed bends. Care was taken to position the objective normal to the sample edge and the camera was aligned axially with the objective in order to minimize coma and off-axis aberration. An attenuator was placed between the collimating and focusing lenses to reduce the peak pixel intensity below the saturation level of the camera.

The beam analyzer displays a calibrated intensity profile in real time by interrogating the camera and capturing the image. The beam analyzer stores the intensity quantized to 8 bits in a 120 × 120 frame array. The pixel-to pixel separation s' in the object plane is $s' = s/M$, where s is the pixel-to-pixel separation on the CCD and the magnification M is set by the objective and the camera-to-sample spacing. By translating a small laser spot, the pixel horizontal center-to-center separation on the CCD was measured to be 19.70 μm and the vertical center-to-center separation was 19.62 μm. The magnification for a sample-to-camera distance of 59.53 cm was $M = 270$. Therefore the center-to-center pixel separation in the object plane was 0.0728 μm horizontally and 0.0726 μm vertically. Also, it had been earlier established that objective lens diffraction effects on the horizontal profile (but not on the vertical profile) are insignificant since the width of the channel (2.1 μm) is much larger than the width of the point spread function distribution in the object plane (0.56 μm).

To obtain the best image, the objective-to-sample distance was adjusted so that the vertical profile on the beam analyzer display was the narrowest possible. A total of eight images were captured with the beam analyzer. To reduce noise and offset effects in the camera, four of these measurements were taken with the laser turned off to act as a background measurement. Signal averaging was achieved by taking the pixel-by-pixel sum of the scans with the laser on and subtracting the pixel-by-pixel sum of the background scans. To accomplish this, each scan was downloaded from the beam analyzer to a computer via an IEEE-488 interface and the result saved on disc for analysis and plotting.

Intensity contour plots for four left-handed bends (Fig. 1) and a straight channel are shown in Fig. 9. The SiO_2 isolation layer is toward the bottom of the plot, air is toward the top, and the outside of each bend is toward the left. Although a bright spot at the sample edge easily identified the measured waveguide, the intensity peak with respect to the channel ridge or sample surface could not be measured precisely and therefore the amount of mode peak shift could not be measured. For this reason, in each plot, the peak in the pixel array has been positioned at $(0, 0)$. Asymmetry in the intensity profiles, seen as a slower decay on the outside of the bend than on the inside of the bend, is negligible for the straight guide (infinite bend radius) and guides with large bend radii but becomes more pronounced as the bend radius decreases. It can also be seen that the mode half-width changes.

Cross-sectional electric field profiles were obtained from the intensity profiles by taking the square root of each intensity data point along the zero vertical position of each curve in Fig. 9. Fig. 10 compares these measured field profiles to analytical profiles calculated using (5) and to BPM results. In order to compare the profiles we have placed the peak of each at $y = 0$. This clearly shows the increased field deformation as the bend radius decreases, with this effect being especially prominent for the 1200 μm bend. There is good agreement between the measured fields and the BPM fields, but the analytical profile does not exhibit the same degree of asymmetry.

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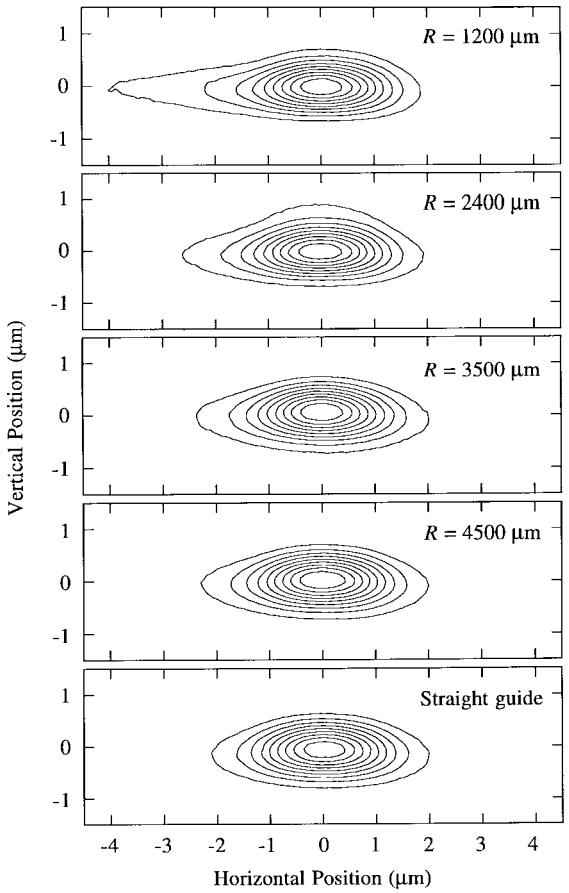


Fig. 9. Contours of mode intensity for a straight waveguide and left-handed bends after 90° of arc propagation. The substrate is toward the bottom of the plots and the outside of the bend is toward the left. The contours shown are from 10% to 90% of the maximum in 10% intervals. The peak of each profile has been placed at $(0, 0)$.

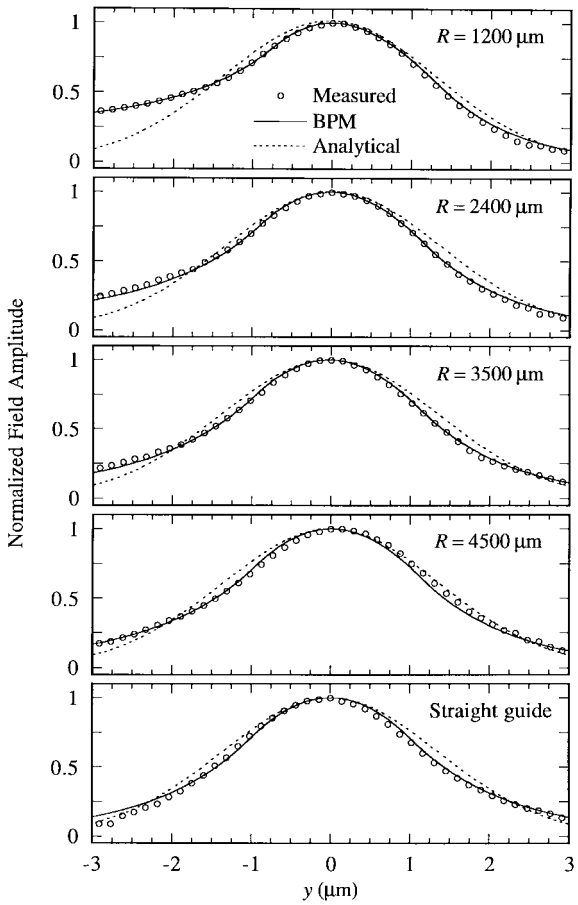


Fig. 10. Comparison of normalized measured, analytical (using (5)), and BPM field profiles for a straight waveguide and left-handed 90° bends after 90° of arc propagation. The outside of the bend is in the $-y$ direction. The peak of each profile has been placed at $y = 0$.

VI. Conclusion

A waveguide sample with a variety of single mode silicon oxynitride waveguide patterns was fabricated and characterized. Loss measurements on U-bends revealed the expected dependence of the bending loss on the bend radius. For bends of radius $R \geq 2000\,\mu$m the measured bending loss agreed well with values calculated from a simple analytical formula. Loss measurements conducted on S-bends revealed the dependence of transition loss on offsets between straight-to-curve and curve-to-curve segments. The transition loss between guides of differing curvature was correctly estimated by an analytical formula derived from a radius dependent, modified Gaussian, waveguide field profile model. The mode field profiles of straight and curved channels were measured using a laser beam analyzer. In curved guides considerable asymmetry in the field profile was observed as the modes shifted toward the outward side of the curvature in both left and right handed bends. The mode field profiles measured in curved guides matched well with 2-D BPM simulated field profiles, but profiles calculated using the radially dependent Gaussian field model did not exhibit the same degree of similarity. Thus, if the field profile is to be accurately modeled, numerical approaches such as BPM are necessary. However, for the goal of obtaining an accurate estimate of transition loss, the simple model works well. Although the field peak shift predicted by this model is 25% greater than that determined by BPM simulations, this difference does not significantly affect the predicted transition loss because of the slow variation with offset of this loss in the vicinity of the optimum offset.

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Vijaya Subramaniam, photograph and biography not available at the time of publication.

Gregory N. De Brabander received the B.S., M.S., and Ph.D. degrees in electrical engineering from the University of Cincinnati, Cincinnati, OH. His Ph.D. dissertation research involved interference-type integrated optical waveguide-micromechanical sensors.

Since 1996, he has been with Hewlett-Packard in San Jose, CA.

Dr. De Brabander received the 1992 Outstanding Graduate Student Research Award in the Department of Electrical and Computer Engineering at the University of Cincinnati.

David H. Naghski (S'80) received the B.A. degree in physics from Franklin and Marshall College, Lancaster, PA, in 1979. He received the M.S. degree in electrical engineering from the University of Colorado, Boulder, in 1985, for his thesis research on deep-level traps in AlGaAs materials. He is pursuing the Ph.D. degree from the Electrical and Computer Engineering Department, University of Cincinnati, Cincinnati, OH. His doctoral research involves mode-field characterization of integrated optical waveguides using near-field scanning optical microscopy.

He spent four years in industry working on electro-optic product development before matriculating in the Ph.D. program.

Mr. Naghski is a member of Sigma Pi Sigma and the Society of Physics Students.

Joseph T. Boyd (S'64–M'69–SM'78) received the B.S. degree in electrical engineering from the Pennsylvania State University, University Park, and the M.S. and Ph.D. degrees in electrical engineering from The Ohio State University, Columbus.

Currently, he is Professor of Electrical and Computer Engineering at the University of Cincinnati, Cincinnati, OH. He has conducted research in the areas of integrated optoelectronics, photonic devices, laser processing, and optical material characterization. Since joining the University of Cincinnati, he has consulted for a number of companies. Prior to this, he was employed by NCR and Rockwell International. He has published over 170 papers and has over 140 conference presentations in the above research areas. In 1988, he served as Guest Editor for a Special Issue on Integrated Optics of the JOURNAL OF LIGHTWAVE TECHNOLOGY. He has presented short-course tutorials on integrated optics and optoelectronics at SPIE meetings and industrial locations. Earlier, he served a six-year term as Secretary of the Administrative Committee (AdCom) of the IEEE Lasers and Electro-Optics Society.

Dr. Boyd is a member of the Optical Society of America, Sigma Xi, Tau Beta Pi, and Eta Kappa Nu.

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